

Linear systems and elimination

Exact extremizers for partial pivoting on orthogonal matrices

Difficulty: challenging **Importance:** interesting to specialist**Status:** Lean verified

Rating rationale (historical): Challenging reflects a global extremal problem with pivot-path constraints in every dimension; specialist impact concerns sharp constants on the orthogonal subclass.

Lean verification — 2026-09-13

The complete negative resolution is now **Lean verified**. The formalization at [lean/](#) exports 17 reviewed declarations, including *NLA.IE05.orthogonalExtremizerConjecture*, which negates the full original universal equality by the exact order-eight witness. It preserves the real orthogonal supremum, positive-diagonal QR convention, all admissible partial pivoting tie paths and the first-available-row rule; the true supremum and the separate asymptotic leading-constant question remain outside scope.

The proof was independently reviewed twice mathematically and independently audited for packaging and remote execution. The authoritative Ubuntu run is [workflow 34751393873, attempt 1](#), with the IE-05 target job [103708463461](#) at immutable candidate revision [71cf72f](#). It accepted all 17 exports through Comparator and Lean’s default kernel, permitted only *propext*, *Classical.choice* and *Quot.sound*, and passed the recorded rejection and sandbox controls. The [permanent verification archive](#) contains the uploaded artifacts and the independent operational seal.

The [source correspondence](#) is preserved from the earlier candidate phase; its “Solved” status and pending-Linux wording are historical and superseded by this verification record.

Mathematical proof and Lean formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, with AI-agent assistance. John Peca-Medlin retains credit for the conjecture and cited element-growth analysis. This is an agent review and kernel-verification record, not external human peer review.

Negative resolution - 2026-09-11

Solved. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, gives an exact order-eight counterexample in the [complete proof, Theorem and Sections 1-4](#). Let $\tilde{L} = L_8 + e_8 e_2^T$, changing only its (8,2) entry from -1 to 0 , and take its positive-diagonal QR factor $\tilde{Q}$. Under the stipulated first-available-row tie rule,

$$\rho_{PP}(\tilde{Q}) = \frac{5272}{63} > \sqrt{\frac{17948132}{2601}} = \rho_{PP}(Q_8).$$

The proof prints exact integer-column descriptions of both orthogonal matrices and verifies all active Schur-complement maxima. This disproves the universal extremizer equality. It does not determine the true orthogonal supremum or refute the separate asymptotic leading-constant conjecture. [Proof PDF](#) · [Standalone TeX](#).

A separate [Codex-agent mathematical review](#) returned **PASS**, with independently written rational checks of both QR conventions, all pivot ties, all 408 active entries and the positive squared growth gap. Substantial ChatGPT/Codex assistance is disclosed; this is automated-agent review, not external human peer review or formal certification. [Authorship](#), [reproducible certificates](#), [source checks](#) and [public-branch audit](#). The original ID, path, statement, historical ratings and earlier checks below are retained. Peca-Medlin retains credit for the conjecture and cited element-growth analysis.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let L_n be the real unit lower triangular matrix whose entries strictly below the diagonal are all -1 . Define Q_n by the unique QR factorization $L_n = Q_n R_n$ with positive diagonal in R_n . For Q_n , use partial pivoting with the first available row chosen in a tie. Is it true that, for every $n \geq 2$,

$$\sup_{Q \in O(n)} \rho_{\text{PP}}(Q) = \rho_{\text{PP}}(Q_n), \quad O(n) = \{Q \in \mathbb{R}^{n \times n} : Q^T Q = I\}?$$

The supremum on the left includes all admissible partial-pivoting paths. The candidate is fully specified by L_n , rather than by an approximate numerical optimizer. This asks for the sharp extremizer, beyond the established exponential order of orthogonal growth.

Reference

Peca-Medlin, *Growth factors of orthogonal matrices and local behavior of Gaussian elimination with partial and complete pivoting*, published in SIAM J. Matrix Anal. Appl. (2024), §3.2 and Appendix B. The paper conjectures this equality and establishes $\rho_{\text{PP}}(Q_n) = 2^{n-1}(1 + o(1))/\sqrt{3}$.

Earlier status check — 2026-09-08

Searches for *GEPP orthogonal conjecture 2026* and the exact paper title found no proof of the extremal equality. The 2026 butterfly paper still identifies orthogonal partial-pivoting growth as open. The August Shah–Urschel results concern different growth questions and pivot strategies; their exponential examples do not establish this exact supremum.

Audit update — 2026-09-10

Rechecked [Peca-Medlin’s manuscript](#), §3.2 and Appendix B, and its [SIAM publication](#), SIAM J. Matrix Anal. Appl. 45 (2024), 1599–1620. The candidate extremizers remain supported by the stated construction and experiments, without a general optimality proof. Searches for subsequent orthogonal GEPP extremizer results found no resolution.